ASL Character Identification

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Introduction and Motivation

ASL is the primary language for ~500K deaf and hard-of-hearing Americans

Automated gesture recognition can bridge communication gaps in real time

Existing systems are often trained on narrow demographics — do they generalize across skin tones, lighting, and hand shapes?

We aim to study whether recognition models are equitable across subjects

Dataset Overview

ASL-HG Dataset from Pranto et al. (2025)

Data covers 36 classes of images (digits 0-10, letters A-Z)

10 subjects in Bangladesh provided 100 images for each class

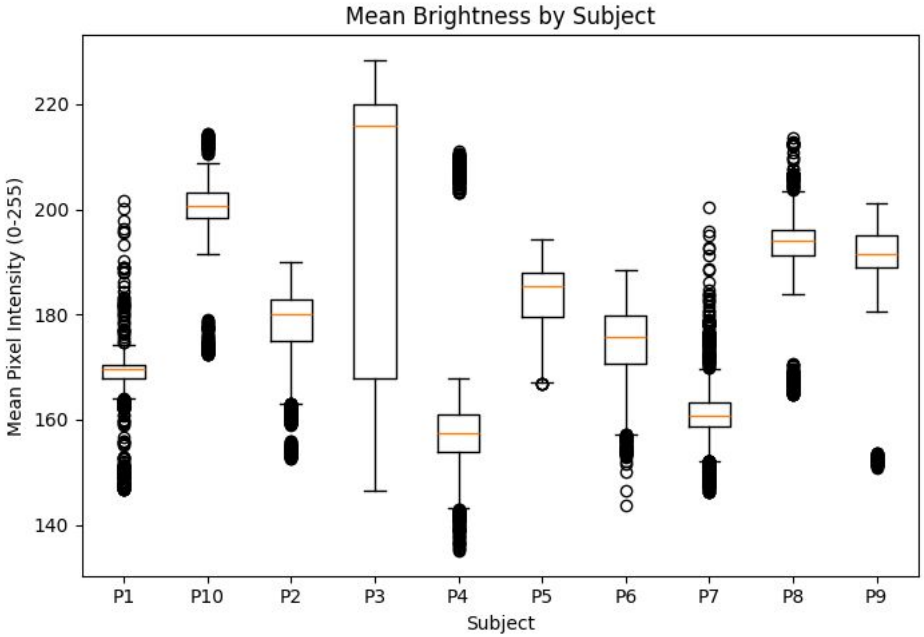
Comes with raw and processed images

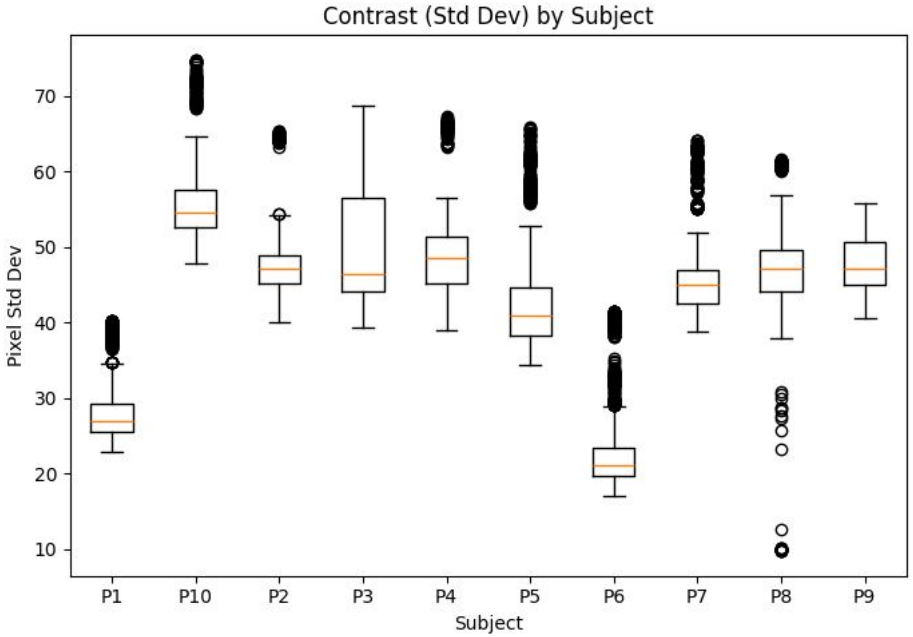
- Raw images are 300 x 300
- Processed images are cropped to different sizes; come with 80-20 train-test split

Data is not uniform, there are different skin tones and lighting conditions across subjects

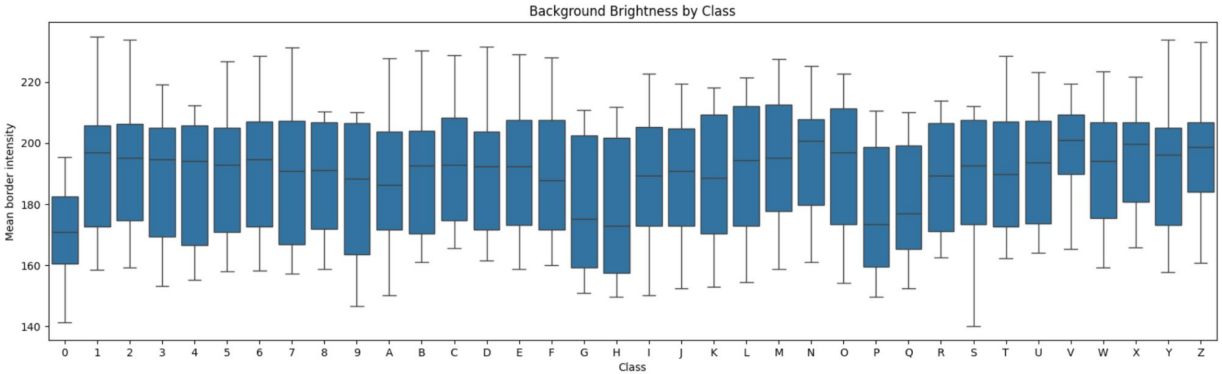
Raw vs Processed ASL Samples

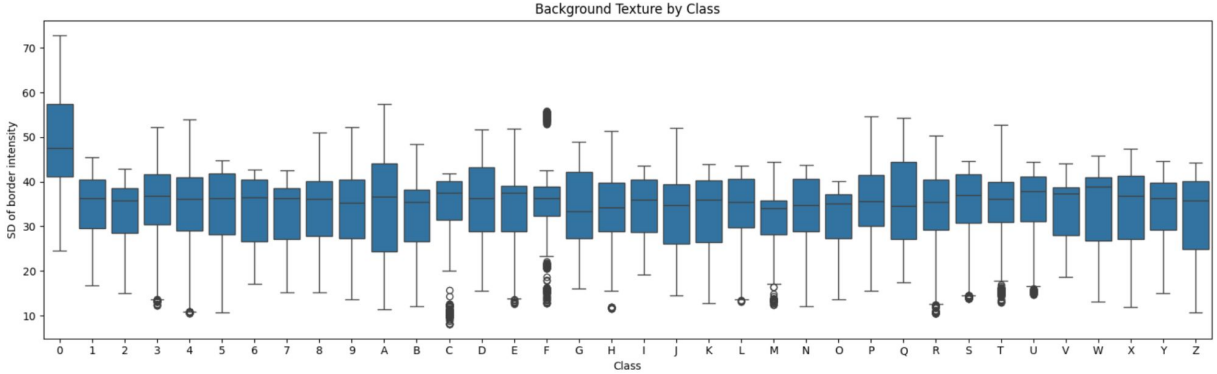
Per Subject Stats





Per Class Stats (Background Border)





Data Leakage

- The provided 80/20 split is NOT subject-level
- The split takes 80 of each subject's 100 images for train, 20 for test
- This means the model sees every hand during training
- We want to test generalization to new people!
- We do: 8 subjects for training (28,800 images), 2 held-out subjects for testing (7,200 images)

Preprocessing Pipeline

- Resize all images to 224×224 (standard input for ResNet, ViT)
- Normalize with ImageNet mean/std (required for pretrained backbones)
- Training augmentation: random rotation ±10°, color jitter (brightness/contrast ±20%)
- Using raw images

Problem Statement

Primary question: How does subject-level variation (skin tone, lighting, hand shape) affect ASL gesture classification accuracy?

Sub-questions:

- Do CNN architectures (ResNet) and Vision Transformers (ViT) differ in cross-subject generalization?
- Which gesture classes suffer the most from subject shift?
- Can a model trained on subjects from Bangladesh accurately classify gestures from subjects with different skin tones (our own test set)?

Future Steps

Build out base identification model through experimentation

Complete analysis on misclassification and identify any potential patterns

Construct out of sample dataset with variety of subjects/lighting conditions

Analyze potential differences in performance in our new dataset

Questions

Any ideas for out of scope models/architecture?

Is the pitched proposal enough to meet the 209B requirements?

What are the next milestones and what is required for each?

Any other suggestions?